



Apple Music gets new features

Apple Music will get: lossless audio support and spatial audio with Dolby Atmos.
<http://dgit.in/jun21-11>



USB C increased max power capacity

USB C goes from 100 W to 240 W support which will enable it to power beefy laptops too. <http://dgit.in/jun21-12>

ASUS ROG MAXIMUS XIII HERO Motherboard

Powering Rocket Lake to the moon

The ASUS MAXIMUS XIII HERO is built for the folks who want to get on the PCIe Gen 4.0 bandwagon, and do it in style. It's the latest motherboard from ASUS for the Intel 11th Gen Rocket Lake desktop processor family and it's built to make the most of it. These new boards are all about bringing stability to those high-performance processors.

BOARD LAYOUT

The board is littered with heatsinks. You've got the usual ones on the VRM circuitry and extends to the I/O shroud. Then there are several NVMe slots on the board which all have heatsinks on them. At a glance, more than half of the motherboard has some heatsink or the other covering it up. Up at the CPU area, we have two metal-clad 8-PIN EPS power connectors for the CPU of which you can boot the motherboard even if just one of them is connected. This powers a 14-phase teamed VRM configuration which uses a Renesas ISL69269 controller in (7+2+1) mode. The power stages used are 14 of Texas Instruments' NexFET 90A which cumulatively provide 1260A of capacity which is quite ridiculous. Then you have four RAM slots using the easy-to-install one-sided clips. They can support up to 128 GB of RAM with speeds of up to 5333 MHz when overclocked. The debug LED, power switch and reset switches are seated on the extreme right edge of the board which makes it quite convenient for overclockers to access. The debug LED's positioning in particular is best in the upper-right corner as it maintains visibility even when installed in a chassis. There are plenty of ARGB and Fan headers on the board. Most of them



Price
₹41,999



DESIGN.....	85
PERFORMANCE.....	90
VALUE FOR MONEY.....	75
FEATURE.....	82

are situated right about the RAM slot with the rest being spread across the bottom edge of the board. There isn't much space for the headers to be distributed across the board.

The MAXIMUS XIII HERO has three x16 PCIe slots of which the first two from the top are directly fed by the processor, so if you use a Rocket Lake processor, you will get access to PCIe Gen 4.0 lanes on those slots. These can run in x16/x0 or x8/x8 or x8/x4 configurations and support multi-GPU setups. The third slot is fed by the Z590 chipset, so you only get PCIe Gen 3.0 lanes. Coming to the M.2 slots on the board, we have four of them, one at the top, two in the middle and one below the second PCIe x16 slot. They're all covered with heatsinks on the top and bottom with thermal pads pre-attached. The first two M.2 slots have PCIe 4.0 x4 lanes from the CPU provided you are running an 11th Gen Core processor. You have to be careful with the number of M.2 SSDs that you connect because there's lane sharing with the PCIe x16 slots. So your PCIe x16 slot will be dropped to x8 lanes if

you use the second M.2 NVMe slot. Read the user manual to understand the various combinations that you can use. Best case is that you can use three of the four M.2 slots if you're running a single graphics card.

For the Audio, we see that ASUS has gone with the Realtek ALC4082 CODEC supplemented with an ESS Sabre 9018Q2C DAC. The capacitors are Chemicon and this configuration should be more than capable to drive high-impedance headphones.

One of the features of Z590 is that it brings native support for Wi-Fi 6E CNVi cards so all premium Z590 boards come with

Wi-Fi 6E already integrated. On the I/O, we see two 2.5G Ethernet ports, 6 USB Type-A 10 Gbps ports, two USB Type-C ports and two USB 2.0 ports. The IGP can only be accessed using an HDMI port.

BIOS

Not much has changed with the BIOS from what we've seen for the past 3-4 generations. For folks wanting to take advantage of the new Intel Adaptive Boost Technology, there's a new toggle under the Extreme Tweaker menu. The rest of the BIOS is pretty solid and provides granular configuration options for the various CPU and board parameters.

VERDICT

ASUS's Z490 boards were more than capable of handling Intel's power hungry 11th Gen Rocket Lake processors so the new Z590 boards are for enthusiasts upgrading from much before. The way the ASUS ROG MAXIMUS XIII is configured, you'll hardly have anything to worry about.

—Mithun Mohandas

SPECIFICATIONS

CHIPSET: Intel Z590 | MEMORY SUPPORT: 4x5300(OC) MHz upto 128GB | LAN: 2x Intel I225-V 2.5G Gigabit Ethernet | AUDIO: ALC 4082 + ESS SABRE9018Q2C | SATA: 6x SATA III | EXPANSION: 3x PCIe X16(x16+x0+x4) | DIMENSIONS (LxWxD): 305 mm x 244 mm | WARRANTY: 3 years.

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